SET THEORETIC COMPOSITION CONTAINMENT FOR THE REDUCED CLOSURE OF THE IMAGE OF J = (T, S) OVER A FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition of the reduced closed subscheme on a closed subset (schemes chapter, Definition schemes-definition-reduced-induced-scheme, canonical tag 01J4; kind definition). cited at statement lines 1886-1888. Defines Z as the reduced closed subscheme whose underlying space is the closure of the image of j.
- The map-groupoids-on-field-image lemma, of which the groupoid case is a special instance (more-groupoids chapter, Lemma lemma-map-groupoids-on-field-image, canonical tag 04Q7). cited at proof lines 1895-1897. One route to the result: the ambient groupoid makes the statement a special case.
- Closure of the image of a product of morphisms over a field equals the product of the closure images (varieties chapter, Lemma varieties-lemma-closure-image-product-map, canonical tag 04Q0). cited at proof lines 1918-1921. Identifies the closure of the image of j_s times _tj with Z_s times_k _tZ, the key input for the containment in Z.

Source reference. Original definition 01J4: sources/stacks-chow/tags.txt, source lines 2001–2001. Cited at statement lines 1886-1888: defines Z as the reduced closed subscheme on the closure of the image of j.

Source reference. Original lemma 04Q7: sources/stacks-chow/tags.txt, source lines 6140–6140. Cited at proof lines 1895-1897: the ambient groupoid (U, U times_S U, pr_1, pr_0, pr_02) makes the statement a special case of this lemma.

Source reference. Original lemma 04Q0: sources/stacks-chow/tags.txt, source lines 6133–6133. Cited at proof lines 1918-1921: the closure of the image of j_s times _tj is Z_s times_k _tZ, the key input for the containment.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume U is the spectrum of a field. Let $Z \subset U \times_S U$ be the reduced closed subscheme (see Schemes, Definition 01J4) whose underlying topological space is the*

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closure of the image of $j = (t, s) : R \to U \times_S U$. Then $pr_{02}(Z \times_{pr_1, U, pr_0} Z) \subset Z$ set theoretically.

Proof. As $(U, U \times_S U, \mathrm{pr}_1, \mathrm{pr}_0, \mathrm{pr}_{02})$ is a groupoid scheme over S this is a special case of Lemma 04Q7. But we can also prove it directly as follows.

Write $U = \mathrm{Spec}(k)$. Denote R_s (resp. Z_s, resp. U_s^2) the scheme R (resp. Z, resp. $U \times_S U$) viewed as a scheme over k via s (resp. $\mathrm{pr}_1|_Z$, resp. pr_1). Similarly, denote ${}_tR$ (resp. ${}_tZ$, resp. ${}_tU^2$) the scheme R (resp. Z, resp. $U \times_S U$) viewed as a scheme over k via t (resp. $\mathrm{pr}_0|_Z$, resp. pr_0). The morphism j induces morphisms of schemes $j_s : R_s \to U_s^2$ and ${}_tj : {}_tR \to {}_tU^2$ over k. Consider the commutative diagram

$$\begin{array}{ccc} R_s \times_k {}_tR & \xrightarrow{c} & R \\ {\scriptstyle j_s \times_t j}\downarrow & & \downarrow{\scriptstyle j} \\ U_s^2 \times_k {}_tU^2 & \longrightarrow & U \times_S U \end{array}$$

By Varieties, Lemma 04Q0 we see that the closure of the image of $j_s \times_t j$ is $Z_s \times_k {}_tZ$. By the commutativity of the diagram we conclude that $Z_s \times_k {}_tZ$ maps into Z by the bottom horizontal arrow. $\square$